PARHAM MORADI

parhammrd.github.io | parhammrd@gmail.com | +1 (639) 470-6551 | Saskatoon, Saskatchewan, Canada

EDUCATION

M.Sc. in Complex Systems & Non-linear Dynamics
Shahid Beheshti University (SBU) Tehran, Iran
2017 – 2020 Overall GPA: 3.19/4

Thesis title: **Station & Detection of Bots in Social Media**
Supervisor: Prof. Gholamreza Jafari

B.Sc. in Physics
Sharif University of Technology (SUT) Tehran, Iran
2012 – 2017 Overall GPA: 2.90/4

Thesis title: **Electronic Structure of Carbon Materials**
Supervisor: Prof. Seyed Akbar Jafari

RESEARCH INTERESTS

- Complex Systems & Network Science
- Computational Social Science & Demography
- Data Science & Analysing Data
- Statistical Physics & Stochastic Processes
- Data Visualization & Quantitative Analysis

EXPERIENCE

Data Science Developer
Cube10
Dec 2023 – Ongoing

- Develop data mining system in a independent micro-service
- Maintenance and review legacy code of ETL data Pipeline

Tech Director
Zamineh Art Platform
Oct 2020 – Dec 2023 Tehran, Iran

- Curating The Art Genome Project for the local art market
- Brainstorm in multi-disciplinary group meeting

Data Science Fellow
Center for Complex Networks & Social Data (CCNSD)
Nov 2019 – Apr 2024 SBU, Tehran, Iran

- Assisted in academic research, including data analysis and theoretical conceptualization
- Managed and curated datasets for researchers, ensuring data quality and consistency

EVENT ORGANIZATION AND LECTURING

- **Organizer & Lecturer of Complex Networks: First Step To Social Media Analysis (2022) at CCNSD**
- **Organizer & Lecturer of First Twitter Data Mining Workshop (2019) at CCNSD**

PUBLICATIONS

Journal Articles

- Parsa Bigdeli Parham Moradi, Reza Jafari (Aug. 2024). "Bots Election: Unveiling the complex network of social bot-nets". In: *PREPRINT (Scientific Reports)*. URL: https://www.researchsquare.com/article/rs-4773878/v1.
- Farzam, Amirhossein et al. (Jan. 2023). "Opinion manipulation on Farsi Twitter". In: *Scientific Reports* 13.333. DOI: 10.1038/s41598-022-26921-5.
- Mohammadi, Saeedeh, Parham Moradi, Andrey Trufanov, et al. (Dec. 2023). "A structural approach to detecting opinion leaders in Twitter by random matrix theory". In: *Scientific Reports* 13.1, p. 21788. ISSN: 2045-2322. DOI: 10.1038/s41598-023-48682-5. URL: https://doi.org/10.1038/s41598-023-48682-5.
- Mohammadi, Saeedeh, Parham Moradi, S. Mahdi Firouzabadi, et al. (July 2022). "The footprint of campaign strategies in Farsi Twitter: A case for 2021 Iranian presidential election". In: *PLOS ONE* 17.7, pp. 1–15. URL: https://doi.org/10.1371/journal.pone.0270822.

Conference Proceedings

- Farzam, A., P. Moradi, S. Mohammadi, et al. (n.d.). "Farsi Twitter as a public sphere: An event-based analysis." In: Oral Presentation at PCNET20 Conference in Rome Italy.
- Farzam, A., P. Moradi, Z. Padar, et al. (n.d.). "Content VS Structure in Farsi Twitter." In: Oral Presentation at Networks 2021 Conference at Indiana University Network Science Institute.

PRESENTATIONS

- **Urgent Path to Data Gathering** *SBU, 2022*
  The lecture at First Step To Social Media Analysis
- **Introduction to Twitter Network & Visualization** *SBU, 2020*
- **Fake News in Social Media** *SBU, 2019*
  The Lecture at First Twitter Data Mining Workshop
- **Spin & Charge Separation in Hubbard Model** *SBU, 2018*
  Seminar & Research Methodology Course

WORKSHOPS & CONFERENCES

- **Sixth IPM Advanced School on Computing & AI**
  Held by School of computer science, Institute for Research in Fundamental Sciences, Summer 2022.
- **Computational Social Science Workshop**
  Held by Students' Union of Mathematical Sciences Sharif University, Winter 2020.
- **School on Theory & Applications of Complex Networks**
  Held by Physics Department, Shahid Beheshti University, Summer 2018.
- **IPM Data Science Day**
  Held by School of Computer Science, Institute for Research in Fundamental Sciences, Fall 2017.
- **21TH School of Advance Physics Education**
  Held by Institute for Advanced Studies in Basic Sciences, Winter 2016.

RESEARCH

Research Assistant in CCNSD

2022 – 2023 SUB, Tehran, Iran

Title: Information Flow & Propagation Modeling

Leader: Prof. Gholamreza Jafari

Description: Studied information spread based on network structure within the political context.

Research Assistant in CCNSD

2021 – 2022 SUB, Tehran, Iran

Title: Monitoring Twitter Activity on Twelve Iran Presidential Election

Leader: Prof. Gholamreza Jafari

Description: Captured social media activity during the presidential election in Iran & prepared Academic Data to initiate research.

Curating TAGP in Zamineh Art Platform

2020 – 2022 Tehran, Iran

Title: Accordance The Art Genome Project in Iran Visual Art

Description: Supported by Zamineh Content Provider Department, besides art student interns, matching the primary Iran art scene with the terms of The Art Genome Project.

Data Moderator in CCNSD

2019 – 2020 SUB, Tehran, Iran

Title: Implement Social Media User Classification

Leader: Prof. Gholamreza Jafari

Supervisor: Dr. Kosar Karimi Pour

Description: Used Botometer to estimate Complete Automation Probability (CAP) distribution on Farsi Twitter through crowdsourcing. Adjusted classification boundaries based on annotations and validated with Kendall's τ.

COMPUTER SKILLS

- Programming Languages Expertise

  python C C++ LaTeX MATLAB Fortran

- Data Analysis & Visualization Libraries

  NumPy pandas nltk SciPy NetworkX TensorFlow matplotlib seaborn

- Tools and Technologies

  Linux Git /(Reg)Ex/ SQLAlchemy

- Database Management Systems

  SQL PostgreSQL mongoDB SQL Server

- Frameworks

  Selenium Scrapy FastAPI

SELECTED COURSES

Physics of Complex Systems	4/4
Application of Computer in Physics, Lab	4/4
Seminar & Research Methodology	4/4
Non-equilibrium Statistical Physics	4/4

ACADEMIC PROJECTS

Final Project in Critical Phenomena Course

Winter 2019 SUB, Tehran, Iran

Title: Transformations of Ising Models

Supervisor: Prof. Seyed Mohammad Sadegh Movahed

Description: Reviewed & reported the transition point with partition function & thermodynamic calculations based on the 1959 Michael E. Fisher article to obtain magnetic susceptibility.

Final Project in Electrodynamics Course

Winter 2018 SUB, Tehran, Iran

Title: Jacobi & Gauss-Siedel to solving Poisson Equation.

Supervisor: Prof. Ali Sadeghi

Description: Used the Jacobi & Gauss-Seidel iteration method in MATLAB to solve numerically Poisson Equation in the square lattice by flat density charge.

Final Project in Condensed Matter Course

Spring 2016 SUT, Tehran, Iran

Title: Spin-Orbit Coupling in Tetragonal Bismuth Bilayer

Supervisor: Prof. Seyed Akbar Jafari

Description: Calculated the band structure of tetragonal bismuth bilayer (TB-Bi), which can be the candidate for topological insulators. Finally, spin-orbit coupling produces a nontrivial topological phase using the tight-binding model approach & the Slater-Koster method.

Project in the Application of Computer in Physics Course

Spring 2015 SUT, Tehran, Iran

Title: Monte Carlo Simulation of The Habard & Ising Model on Honeycomb Lattice

Supervisor: Prof. Seyed Akbar Jafari

Description: Used Fortran to simulate & briefly study the model and using different computational techniques.

Collaborate in a Project

Winter 2015 SUT, Tehran, Iran

Title: Flat Histogram on Ising Model

Supervisor: Prof. Mohammad Reza Ejtehadi

Description: Studied the flat histogram method to decrease computational cost in solving the Ising model base on the Monte Carlo method. The final result as C++ program ran with cluster calculation.